



ATTransUNet: An enhanced hybrid transformer architecture for ultrasound and histopathology image segmentation

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ABSTRACT

Recently, researchers have introduced Transformer into medical image segmentation networks to encode long-range dependency, which makes up for the deficiencies of convolutional neural networks (CNNs) in global context modeling, and thus improves segmentation performance. However, in Transformer, due to the heavy computational burden of paired attention modeling between redundant visual tokens, the efficiency of Transformer needs to be further improved. Therefore, in this paper, we propose ATTransUNet, a Transformer enhanced hybrid architecture based on the adaptive token for ultrasound and histopathology image segmentation. In the encoding stage of the ATTransUNet, we introduced an Adaptive Token Extraction Module (ATEM), which can mine a few important visual tokens in the image for self-attention modeling, thus reducing the complexity of the model and improving the segmentation accuracy. In addition, in the decoding stage, we introduce a Selective Feature Reinforcement Module (SFRM) to reinforce the representation of and attention to key tissues or pathological features. The proposed ATTransUNet is evaluated on the basis of three medical image segmentation datasets. The results show that ATTransUNet achieves the best segmentation performance compared with the previous state-of-the-art models, and the proposed method is also competitive in terms of the network parameters and computation.

1. Introduction

The design of accurate, fast, and robust medical image segmentation algorithms has always been one of the major challenges in the field of computer-aided diagnosis. Segmentation of organs or diseased areas from medical images is of great significance for predicting disease progression, developing treatment strategies and patient prognosis.

Early medical image segmentation algorithms such as statistical models, contour-based methods, and machine learning-based algorithms are generally used [1–3], but the segmentation algorithms are imprecise. With the development of convolutional neural networks (CNNs) in computer vision, CNNs are rapidly used in medical image classification and segmentation tasks. U-Net [4], the early medical image segmentation convolutional network, consists of a symmetric encoder-decoder with a skip connection, which can fuse image features of different levels and significantly increase the accuracy of segmentation. This symmetrical structure has become the mainstream of

medical image segmentation networks [5–9]. They achieve outstanding performance on many challenging data sets and prove the effectiveness of CNNs in learning and representing features to segment the organ or lesion from medical images.

However, an obvious limitation of a convolutional neural network (CNN) is the lack of the ability to model long-range dependency [10–12], which refers to the correlation between distant pixels in an image. In CNN, due to the limitation of the receptive field size, each convolutional kernel only focuses on the local area of the image, rather than the global context. Some works have been taken to address these issues by utilizing dilate convolution [13], feature pyramid [14], multiscale fusion [15,16], and self-attention mechanism based on the convolution feature [17,18]. However, these methods only extend the receptive field of the network, but still cannot model the global context of the features well, especially when dealing with segmentation scenes with complex and changeable pathological features and tissue structure in

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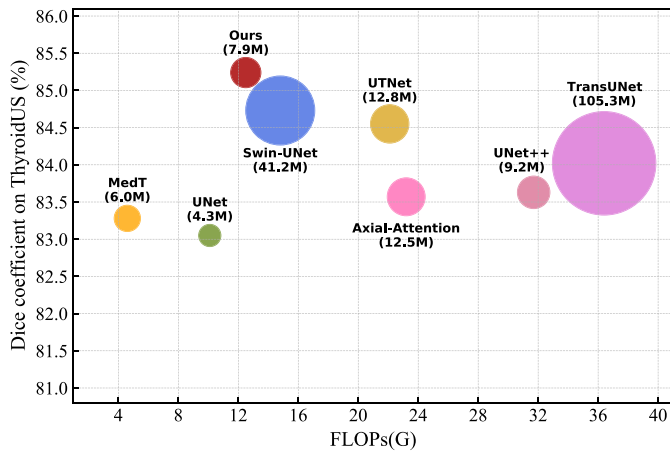


Fig. 1. Dice coefficient on ThyroidUS dataset based on previous state-of-the-art CNN and Transformer models. Our proposed ATTransUNet outperforms all the baselines, and our method is also competitive in network parameters(M) and FLOPs(G).

medical images. For example, to accurately locate and segment thyroid nodules in a thyroid ultrasound image, it is necessary to establish long-range dependency between nodules and surrounding soft tissue structures (such as the trachea, artery, hyoid muscle, skin, etc.), but the methods mentioned above cannot accomplish this task.

The self-attention mechanism of the Transformer [12] can capture the long-range dependency of different features, which makes up for the deficiency of CNN in global context modeling. Recently, the Transformer-based vision model [11,19–21] has been pre-trained on large-scale data sets by using tokenized image patches, surpassing CNN in image classification. Besides, Transformer-based architectures have also been studied for semantic segmentation [22–25]. In medical image segmentation, several medical image segmentation works with Transformer or hybrid architecture have emerged recently, such as TransUNet [26], MedT [27], Swin-UNet [28], UTNet [29], etc.

Although the Transformer based medical image segmentation networks have improved segmentation performance, training and deployment of such methods still have the following limitations:

- (a) Redundancy of computation. The above segmentation methods implement the self-attention mechanism in the full image or local window of the feature map, and the computational complexity has a quadratic relationship with the map or window size. This leads to redundant computing and high inference costs, which hinders their running in real-time.
- (b) The traditional Transformer has no inductive bias [30,31], so cannot distinguish the degree of importance of features on the limited data. In the decoding stage, the above segmentation network's ability to represent important pathological features or tissue structure is weak.

Based on the above analysis, We propose ATTransUNet, a medical image segmentation framework. ATTransUNet is based on an encoder-decoder structure and uses Transformer/CNN hybrid architecture. The contributions of this paper can be summarized as follows:

- (1) The Adaptive Token Extraction Module (ATEM) is proposed. Instead of relying on tokens formed by uniform splitting, this module is able to mine a few tokens representing aggregated semantics in the feature map, and Transformer models these tokens. Our method reduces the computation load and enables the network to have global context modeling ability. Furthermore, we propose the Feature Reprojection Mechanism to remap tokens to feature maps for up-sampling of segmentation.

- (2) The Selective Feature Reinforcement Module (SFRM) is proposed. SFRM models pixel certainty in feature space to selectively encourage the network to focus on the representation of major semantic features. With SFRM, the learning center of the network can always stay in the tissue areas with contextual information, which improves the ability of the network to represent pathological tissues or structures.

We take the thyroid ultrasound (ThyroidUS) dataset collected from Tianjin Medical University Cancer Institute and Hospital as an example, Fig. 1 shows that the method proposed in this paper has achieved the best performance in segmentation accuracy compared with other baselines. At the same time, our method is competitive in terms of computation (FLOPs) and parameters (Params).

2. Related work

2.1. Segmentation methods based on CNN

CNN models have been widely studied and applied in medical image segmentation. U-Net [4], a typical U-shaped network consisting of a symmetrical encoder-decoder with skip connections, has become the choice for medical image analysis. Subsequently, researchers proposed a number of variants based on U-Net architecture. ResUNet designed by Xiao et al. [5] introduced residual connection in the encoder to achieve feature enhancement for deep semantics. UNet++ presented by Zhou et al. [32], and UNet3+ presented by Huang et al. [33], introduced dense connection at the position of skip connection to capture multi-scale features and balance the semantic gap between encoder and decoder. Although CNN has made great progress in medical image segmentation, its performance is generally unsatisfactory due to the inherent limitations of convolution operations, especially for target structures that exhibit large inter-patient differences in texture, shape and size [26].

To improve its function in the global context modeling, some studies have attempted to integrate the self-attention mechanism into the CNN. The main purpose is to attach the attention weights in terms of channels or spatial. Axial-Attention [18] proposed a Gated Axial-Attention module to increase the weight of features. Wang et al. [34] developed a non-local operator and inserted it into the convolution layer to model the long-range dependency of deep semantics. Although these strategies can improve the ability to distinguish semantic features from context. The ability to extract long-range semantic information is still very limited due to the single embedded form of the self-attention module.

2.2. Segmentation methods based on transformer

Vision Transformer (ViT) [11] achieves excellent classification performance in image classification. The general steps were as follows: the input images were tokenized and encoded with position embeddings, then a series of Transformer layers were used for self-attention modeling, and finally the network outputs classification tokens. Later, researchers proposed more efficient variants of ViT, such as DeiT [35], Swin [36], LeViT [21], TokenLearner [37], etc.

Due to Transformer's success in computer vision, a new paradigm in medical image segmentation was formed. TransUNet [26] is the first medical image segmentation model combining Transformer and CNN. Later, MedT [27] introduced a Gated Axial-Attention module to alleviate the problem of a lack of datasets in medical imaging. Swin-UNet [28] proposed the first U-shaped segmentation architecture entirely based on Transformer while introducing shift-window (Swin) attention to reduce computational redundancy. UTNet [29] maps the *Key*, *Value* of self-attention to the smaller size and achieves better context modeling ability and segmentation effect. However, the above methods have obvious limitations: although some studies try

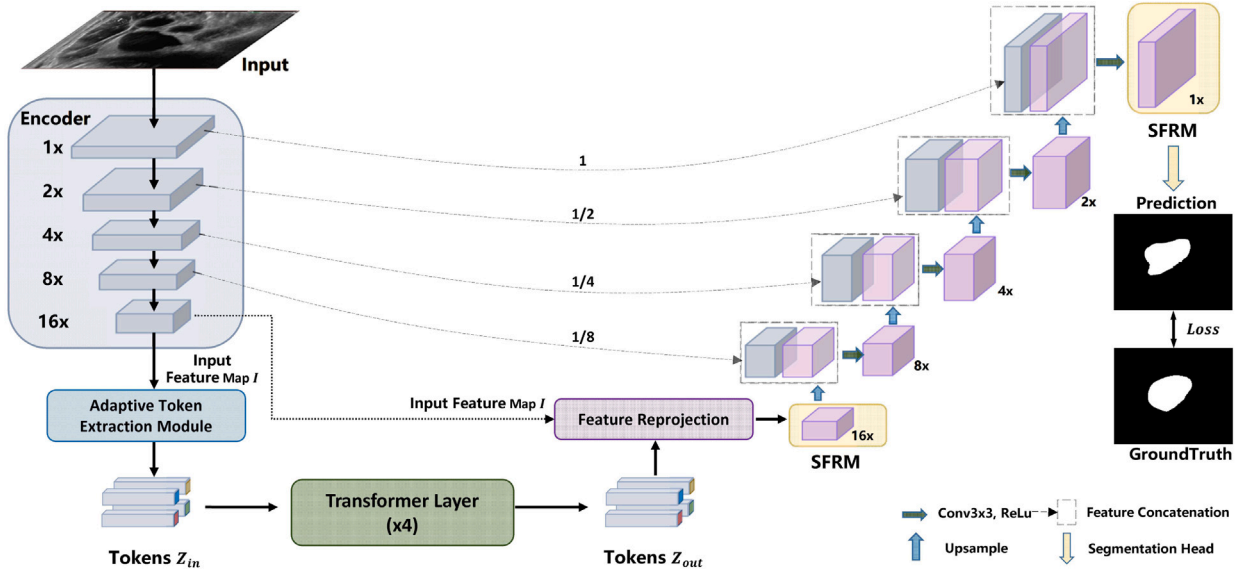


Fig. 2. Overview of the network architecture. ATTransUNet is based on an encoder-decoder structure. The number of basic channels in the first layer of the encoder is set to 32 (1x) and the number doubles in each of the four subsequent down-sampling layers. At the end of the encoder, Adaptive Token Extraction Module (ATEM) is used to generate tokens for input to Transformer. Feature Reprojection remaps the output tokens to the feature map. In the decoder, the Selective Feature Reinforcement Module (SFRM) is applied to select and enforce features, and the prediction is output by four times up-sampling.

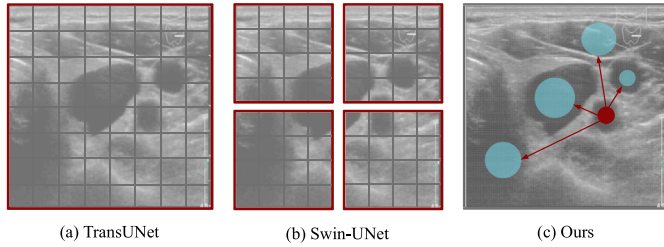


Fig. 3. Comparison of token generation methods of different models. (a) TransUNet divides the image into small patches as tokens for input to Transformer. (b) Swin-UNet and UNet divide the image into patches as tokens, and feed the tokens of shift or local windows into Transformer. (c) Our method aggregates local tissues or features into tokens, as the area covered by bubbles, for input Transformer.

to introduce different methods (such as Swin Attention) to reduce the computational complexity, the complexity has a quadratic relation with input window size. The networks cannot make full use of the contextual information obtained through self-attention modeling in the up-sampling stage.

3. Methods

In Section 3.1, we will introduce the overall network architecture. In Section 3.2, we will introduce the principle of the **Adaptive Token Extraction Module (ATEM)**. In Sections 3.3 and 3.4, we will elaborate on the motivation and details of the **Feature Reprojection Mechanism** and **Selective Feature Reinforcement Module (SFRM)**. In Section 3.5, we compare the computational complexity of self-attention modeling for our method with other methods, and will provide a detailed analysis.

3.1. Network architecture

Fig. 2 shows the architecture of ATTransUNet proposed in this paper. In general, the network consists of a standard U-shaped encoder and decoder.

For a given input image, The deep feature representation is encoded by four times down-sampling. The number of basic channels in the first

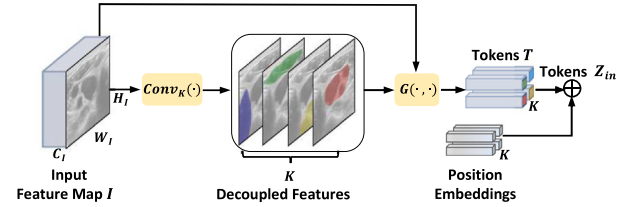


Fig. 4. Adaptive Token Extraction Module. The input feature map I is decoupled to generate K separate semantic features. Aggregate K separate features with feature map I to generate tokens.

layer of the encoder is set to 32 and the number doubles in each of the four subsequent down-sampling layers, where each down-sampling layer contains maximum pooling and a pair of convolution operations. At the end of the encoder, the feature map is converted into a group of tokens through the **Adaptive Tokens Extraction Module**, which is input into Transformer layers to model the context relations of high-level semantic concepts. For Transformer layers, the number of layers is set to 4, and it is composed of four heads of self-attention and a standard MLP layer. Layer normalization is adopted in the self-attention module. Furthermore, the tokens output by Transformer are reprojected back to the spatial feature maps by **Feature Reprojection Mechanism** to be used for up-sampling. In the decoder, the **Selective Feature Reinforcement Module** is applied to the feature maps of the first and the last convolution layer, so as to enhance the semantic features in the image from coarse-grained and fine-grained. The segmentation results are output by four times up-sampling.

3.2. Adaptive token extraction module

Fig. 3 qualitatively compares the differences between the previous segmentation networks (TransUNet [26], Swin-UNet [28], UNet [29], etc.) and ATTransUNet in self-attention modeling. In previous methods, an image was divided into patches (e.g., 16×16) as tokens. Transformer calculated self-attention for all tokens globally or within a window, and the complexity was in a quadratic relationship with the resolution of the feature map or window. However, self-attention was essentially low in rank for long sequences [29,38]. In an image,

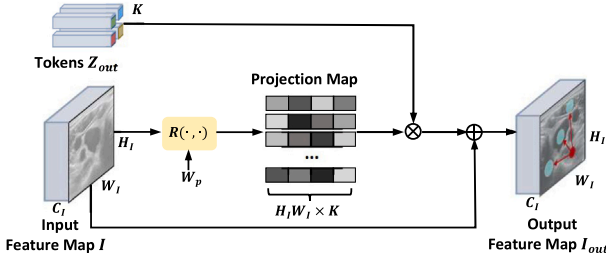


Fig. 5. Feature Reprojection Mechanism remaps tokens to feature map, to fuse with contextual information across the semantic region.

most pixels in the local area had similar semantics. Thus, computing self-attention for all pixels of the feature map was inefficient and redundant.

Based on the above analysis, and inspired by work [37,39], We propose an **Adaptive Token Extraction Module (ATEM)** that can mine a small number of tokens representing important semantics in the image. Fig. 3(c) illustrates how tokens are generated from ATEM. For the input feature map, ATEM aggregates local tissues or features into tokens, as the area covered by bubbles in Fig. 3(c), to be used for self-attention modeling.

Fig. 4 shows an implementation of the **Adaptive Tokens Extraction Module**. For the feature map $I \in \mathbb{R}^{H_I \times W_I \times C_I}$ generated by the encoder (H_I and W_I are the height and width, and C_I is the number of channels). Low-dimensional embedding and feature decoupling are performed by convolution operation on feature map I to extract the K separate semantic features, which are expressed as follows:

$$A = \text{Conv}_K(I) \quad (1)$$

$\text{Conv}_K(\cdot)$ is the convolution operation of K channels, and $A \in \mathbb{R}^{H_I \times W_I \times K}$ represents the K feature maps obtained by decoupling. A is activated and the $H_I W_I$ dimension is flattened. Matrix multiplication is carried out with $H_I W_I$ tokens of feature map I , and the group T consisting of tokens is formed, which is expressed as follows:

$$T = G(A, I) \\ = \sigma(\text{flatten}_{H_I W_I}(A))^T \otimes \text{flatten}_{H_I W_I}(I) \quad (2)$$

Where $\sigma(\cdot)$ represents *Sigmoid* function, $\otimes$ represents matrix multiplication, and $T \in \mathbb{R}^{K \times C_I}$ is composed of the token t_i :

$$T = [t_1, \dots, t_i, \dots, t_K]^T, t_i \in \mathbb{R}^{C_I}, i \in \{1, \dots, K\} \quad (3)$$

K is the number of tokens, C_I is the dimension of tokens. In general, K is much smaller than $H_I W_I$.

After tokenization of the feature map, T is encoded with the relative position embeddings [11] to learn the context of different aggregated features, which is expressed as follows:

$$Z_{in} = T + E_{pos} = [t_1, \dots, t_i, \dots, t_K]^T + E_{pos} \quad (4)$$

$E_{pos} \in \mathbb{R}^{K \times C_I}$ are the learnable position embeddings, and z_i represents token embedded with position information that form $Z_{in} \in \mathbb{R}^{K \times C_I}$:

$$Z_{in} = [z_1, \dots, z_i, \dots, z_K]^T, z_i \in \mathbb{R}^{C_I}, i \in \{1, \dots, K\} \quad (5)$$

For Z_{in} , we use standard Transformer layers to model self-attention.

3.3. Feature reprojection mechanism

Segmentation tasks require pixel details, but tokens used for global context modeling are difficult to preserve such details. As a supplement and extension of the Adaptive Tokens Extraction Module, tokens output by Transformer layers are fused with input feature map I and they

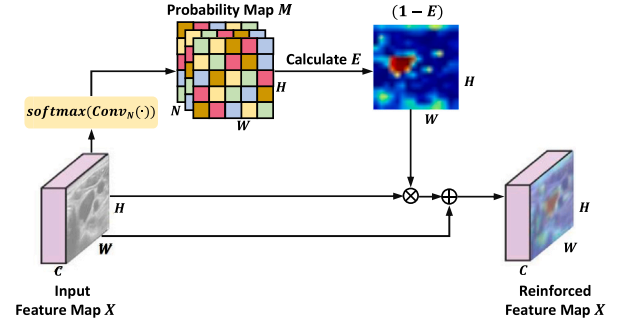


Fig. 6. Selective Feature Reinforcement Module (SFRM). SFRM selects features for reinforcement through pixel certainty.

are reprojected to spatial feature maps to be used for up-sampling. The process is expressed as follows:

$$I'_{out} = R(I, W_p) \otimes Z_{out} \\ = \sigma(\text{flatten}_{H_I W_I}(I) \otimes W_p) \otimes Z_{out} \quad (6)$$

$$I_{out} = \text{reshape}_{H_I W_I}(I'_{out}) + I \quad (7)$$

Here, $R(\cdot, \cdot)$ represents the *Reprojection* function, $\sigma(\cdot)$ represents the *Sigmoid* function, $I \in \mathbb{R}^{H_I \times W_I \times C_I}$ as the feature map before the input Adaptive Token Extraction Module, $W_p \in \mathbb{R}^{C_I \times K}$ as the learnable linear projection matrix, and $Z_{out} \in \mathbb{R}^{K \times C_I}$ as the tokens of the output of Transformer layer.

The **Feature Reprojection Mechanism** is essentially a pixel-by-pixel query calculation, as shown in Fig. 5. Through $R(\cdot, \cdot)$, channels of feature map I is reprojected to K -dimension and normalized to form a **Projection Map** with size $H_I W_I \times K$. Each K -dimension embeddings of the **Projection Map** correspond to the attention weight of K tokens in Z_{out} . The **Projection Map** is multiplied by Z_{out} to reproject tokens back to the feature map, and the pixels of the reprojected feature map are fused with contextual information across the semantic region.

3.4. Selective feature reinforcement module

The Transformer enables global context modeling so that semantic features can obtain contextual information from different locations. To keep the network focused on the areas of the focal tissue, we introduce the **Selective Feature Reinforcement Module (SFRM)**.

Specifically, as shown in Fig. 6, the dimension of the feature maps $X \in \mathbb{R}^{H \times W \times C}$ from the convolution layer is mapped to $H \times W \times N$ through $1 \times 1 \times N$ convolution kernels, and the probability values of the pixels are normalized along the channels N by the *softmax* function. Where N represents the number of segmentation categories of segmentation tasks, which is shown as follows:

$$M = \text{softmax}(\text{Conv}_N(X)) \quad (8)$$

$\text{Conv}_N(\cdot)$ represents the convolution operation of N channels, and the **Probability Map** $M \in \mathbb{R}^{H \times W \times N}$. In the feature map, the weight value $E_{i,j}$ of each pixel is measured by calculating the information entropy of the $M_{i,j} \in \mathbb{R}^{1 \times 1 \times N}$ of each pixel in M .

$$E_{i,j} = \sum_{n=1}^N -\frac{M_{(i,j),n} \cdot \log(M_{(i,j),n})}{\log(N)} \quad (9)$$

When the class probabilities of pixels tend to a uniform distribution, the larger the entropy is, the more likely it is to belong to uncertain edge or texture features. However, when the class probability of pixels tends to a certain category, the entropy is small and the semantic distribution is unified. The pixels are more likely to belong to the regular tissues or feature regions. **SFRM** enables the network to adaptively reduces

the focus on tissues and pathological features of the edge and enhance major semantic features, and its formula is defined as follows:

$$X = X + (1 - E) \cdot X \quad (10)$$

In the above formulas, $E \in \mathbb{R}^{H \times W \times 1}$ is formed by weights values of pixels.

3.5. Complexity analysis

In this section, we compare the complexity of the token mining method proposed in this paper in self-attention modeling with the previous approach. In the Transformer, multihead self-attention jointly inferences attention from different representation subspaces, which are then transformed by a feedforward network (FFN) to establish global context. For simplicity, this section analyzes the complexity of all methods based on single-head self-attention. TransUNet [26] calculates self-attention based on the full map. For input features $Z_I \in \mathbb{R}^{H_I W_I \times C_I}$, three linear layers are used to obtain embeddings $Q_I, K_I, V_I \in \mathbb{R}^{H_I W_I \times C_h}$. (C_h is the embedding dimension of a single-head). The self-attention formula is defined as follows:

$$\text{Attention}(Q_I, K_I, V_I) = \text{softmax}\left(\frac{Q_I \otimes K_I^T}{\sqrt{C_h}}\right) \otimes V_I \quad (11)$$

The operation of the $H_I W_I \times C_h$ size matrix results in the complexity of $O((H_I W_I)^2 C_h)$, which makes it difficult to apply to feature map with large resolution. Swin-UNet [28] and UNet [29], etc. divide input feature Z_I into smaller local windows of size $M_I \times M_I$ and model the self-attention on windows. Its computational complexity is reduced to $O(M_I^2 H_I W_I C_h)$ [28,29].

In our method, for the input $Z_{in} \in \mathbb{R}^{K \times C_I}$, transform the embeddings $Q_I, K_I, V_I \in \mathbb{R}^{K \times C_h}$, and the Transformer layers model self-attention to K tokens. According to the formula (6) and (2), the complexity is $O(K^2 C_h)$ and $O(K H_I W_I C_I)$. formula (2) is only used to pre-process tokens but is not involved Transformer modeling, so its complexity is negligible. As K is much smaller than $H_I W_I$, the complexity is exponentially reduced compared with TransUNet, especially in multilayer Transformer layers. Compared with Swin-UNet and UNet, etc, our method has less amount of computation.

4. Experiments and results

4.1. Experimental setup

4.1.1. Datasets

The study has been approved by the Medical Ethics Committee of Tianjin Medical University Cancer Institute and Hospital. Written consent has been obtained from each patient after a full explanation of the purpose and nature of all procedures used. In this paper, the thyroid ultrasound (ThyroidUS) dataset, Gland segmentation (GlaS) dataset [40], and MoNuSeg cell segmentation dataset [41] are used to evaluate the performance of the proposed method. The ThyroidUS contains 1415 ultrasound scan images for training and 327 images for testing; The GlaS contains 85 Gland microscopic staining images for training and 80 images for testing. The MoNuSeg dataset contains 30 cell images for training and 14 images for testing.

4.1.2. Settings and evaluation metrics

All experiments in this paper were based on an NVIDIA Titan V (12 GB) GPU, and were coded by using the open-source framework PyTorch. The batch size was set to 8. The SGD optimizer was adopted, in which the momentum and weight decay were set as 0.9 and 1e-4 respectively, and the initial learning rate was set as 0.01. The CosineAnnealingLR was used to train 200 epochs for the model. For all baselines in this paper, the same input size and random enhancement strategy (horizontal flip, vertical flip, etc.) were used for images. All the baseline models in the manuscript for comparison and our model were trained

several times (three times) to average the accuracy over the test set to avoid errors due to random parameter initialization. To further illustrate the persuasiveness of the experimental results, we provide the p -values between the results of other models and our model to show the significant differences between the result pairs. The specific p -values are shown in Table 1.

The sum of Cross Entropy loss and Dice loss was used as the global loss function for network training. Dice coefficient (%) and IoU (%) were used to evaluate the segmentation accuracy of different baseline models in corresponding data sets. In addition, to adapt to the real-time performance of medical-assisted diagnosis scenes, the computation (FLOPs) of the model and the number of network parameters (Params) were also used as evaluation indexes for comparative experiments.

4.2. Comparison with state-of-the-art methods

The proposed model is compared with seven state-of-the-art methods on different datasets. Specifically, the seven methods can be divided into three categories, described in detail as follows:

- **Methods based on CNNs:** UNet [4] and UNet++ [32] are selected as the representatives of CNNs to compare with the network proposed in this paper. UNet is composed of an encoder-decoder with a skip connection; UNet++ introduces a dense connection at the skip connections position in the standard UNet structure.
- **Methods based on Self Attention:** Axial-Attention [18], as a segmentation task, combines the self-attention with the CNN and introduces the Gated Axial Attention after the convolutional operation. The structure set in [27] is adopted as one of the baseline models for comparison in this paper.
- **Methods based on global Transformer:** TransUNet [26], MedT [27], UNet [29], and SwinUNet [28] are selected as the medical segmentation networks with Transformer architecture to compare with our model. TransUNet is the first medical image segmentation network combining Transformer structure with CNN. MedT proposed a Gated Axial-Attention mechanism based on Transformer. UNet and SwinUNet use a more efficient self-attentional modeling approach. All the above baseline models have adopted their official versions.

As illustrated in Table 1, our proposed network achieves better results than the other methods. In terms of Dice and IoU, Our model achieves the best results on all three datasets. In terms of Dice, our model has improved by 0.51%, 0.35% and 0.52%, respectively, compared with the best result achieved by Swin-UNet on three datasets. In terms of IoU, there is an improvement of 0.58%, 0.53% and 0.56% compared with Swin-UNet on three datasets. Compared with the worst-performing model, our model achieves an improvement of 2.19% and 2.67% regarding to Dice and IoU on ThyroidUS dataset, respectively. UNet++ is slightly better than UNet due to the introduction of dense connections to balance the differences in features. One fact is that all the Hybrid Transformer Architecture except MedT achieves better results than CNN models, which is attributed to the Transformer's ability of global modeling. However, UNet has a smaller number of parameters than our proposed ATTransUNet. This is because our model develops from UNet, so this will definitely lead to an increase in the number of parameters. But UNet's accuracy is lower in Dice and IoU than ours, which is obviously not suitable for the high-accuracy medical segmentation task. Similarly, our model is also not optimal in terms of FLOPs, but considering the trade-offs between accuracy and the complexity of the model, ATTransUNet is more competent in the medical segmentation field. Finally, we show the p -values between each of these pairs of results for them to explain the statistical difference in the results.

Table 1
Comparison with state-of-the-art segmentation methods on ThyroidUS, GlaS and MoNuSeg dataset.

Methods	ThyroidUS		GlaS		MoNuSeg		Params (M)	FLOPs (G)
	Dice	IoU	Dice	IoU	Dice	IoU		
UNet	83.05 _{<i>p</i>=0.005}	76.87 _{<i>p</i>=0.006}	87.26 _{<i>p</i>=0.002}	79.13 _{<i>p</i>=0.001}	77.52 _{<i>p</i>=0.003}	63.68 _{<i>p</i>=0.004}	4.3	10.1
UNet++	83.63 _{<i>p</i>=0.021}	78.05 _{<i>p</i>=0.017}	87.98 _{<i>p</i>=0.011}	79.61 _{<i>p</i>=0.001}	77.53 _{<i>p</i>=0.010}	63.52 _{<i>p</i>=0.006}	9.2	31.7
Axial-Attention	83.57 _{<i>p</i>=0.003}	77.30 _{<i>p</i>=0.003}	88.12 _{<i>p</i>=0.002}	79.69 _{<i>p</i>=0.001}	77.80 _{<i>p</i>=0.012}	63.86 _{<i>p</i>=0.016}	12.5	23.2
TransUNet	84.02 _{<i>p</i>=0.011}	78.13 _{<i>p</i>=0.020}	88.25 _{<i>p</i>=0.007}	80.40 _{<i>p</i>=0.003}	78.30 _{<i>p</i>=0.045}	64.41 _{<i>p</i>=0.029}	105.3	36.4
MedT	83.28 _{<i>p</i>=0.002}	77.21 _{<i>p</i>=0.003}	86.37 _{<i>p</i>=0.001}	77.95 _{<i>p</i><0.001}	78.17 _{<i>p</i>=0.036}	64.29 _{<i>p</i>=0.032}	6.0	4.6
UTNet	84.55 _{<i>p</i>=0.090}	78.57 _{<i>p</i>=0.056}	88.65 _{<i>p</i>=0.043}	80.89 _{<i>p</i>=0.038}	78.36 _{<i>p</i>=0.044}	64.83 _{<i>p</i>=0.071}	12.8	22.1
Swin-UNet	84.73 _{<i>p</i>=0.129}	78.96 _{<i>p</i>=0.181}	89.28 _{<i>p</i>=0.199}	82.02 _{<i>p</i>=0.140}	78.64 _{<i>p</i>=0.076}	64.95 _{<i>p</i>=0.120}	41.2	14.8
Ours	85.24	79.54	89.63	82.55	79.16	65.51	7.9	12.5

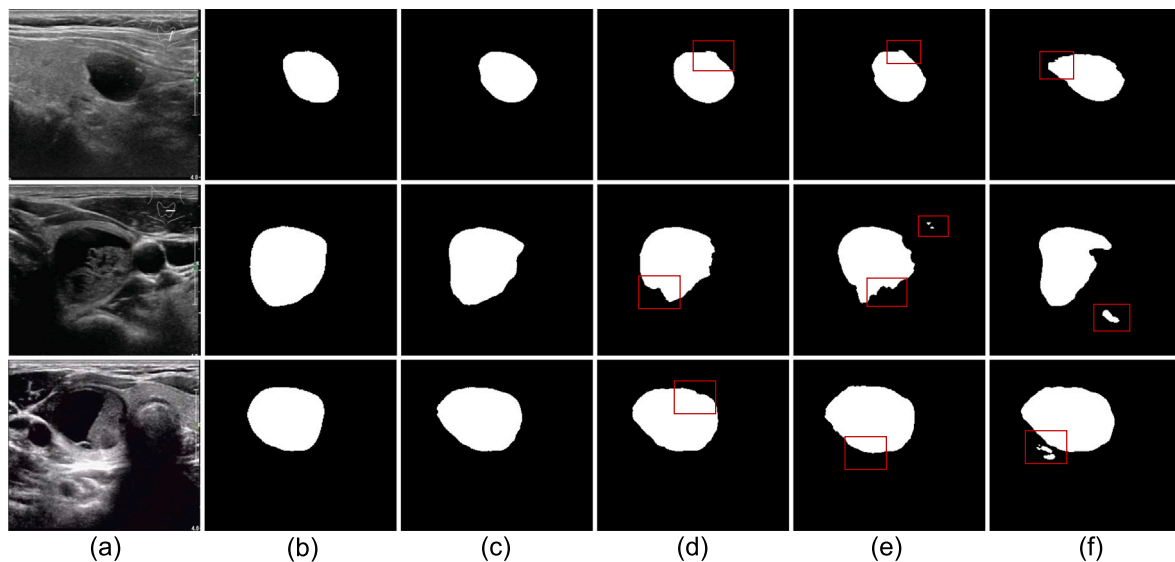


Fig. 7. Example segmentation results of the state-of-the-art methods on ThyroidUS: (a) the original images; (b) ground truths; (c) the results of ATTransUNet; (d) the results of Swin-UNet; (e) the results of TransUNet; (f) the results of UNet.

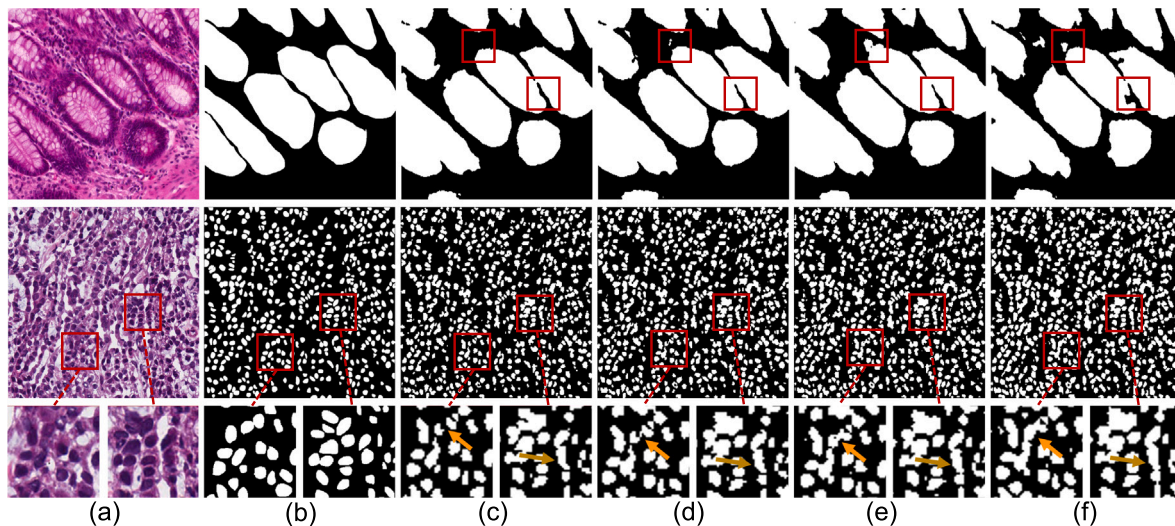


Fig. 8. Example segmentation results of the state-of-the-art methods on GlaS (top) and MoNuSeg (bottom): (a) the original images; (b) ground truths; (c) the results of ATTransUNet; (d) the results of Swin-UNet; (e) the results of TransUNet; (f) the results of UNet.

Fig. 7 shows the segmentation effect of our model on the ThyroidUS datasets. Fig. 7(a)–(b) represents the original images and ground truths respectively, and Fig. 7(c)–(f) represents the segmentation results of ATTransUNet, Swin-UNet, TransUNet and UNet on the ThyroidUS dataset. It can be observed that ATTransUNet is more accurate in locating thyroid nodules, and its segmentation results are closest to the ground truth. Moreover, ATTransUNet retains more detailed shape information

than other models. It is worth noting that in Fig. 7(e) and Fig. 7(f), the previous methods show mis-segmented regions, that is, the area outside the thyroid nodule is incorrectly segmented as nodular tissue, while this situation does not occur in ATTransUNet. This is due to ATTransUNet's ability to establish long-range dependency on tissues and structures at different locations. At the same time, the representation of these features is enhanced in the up-sampling stage of the model, so that

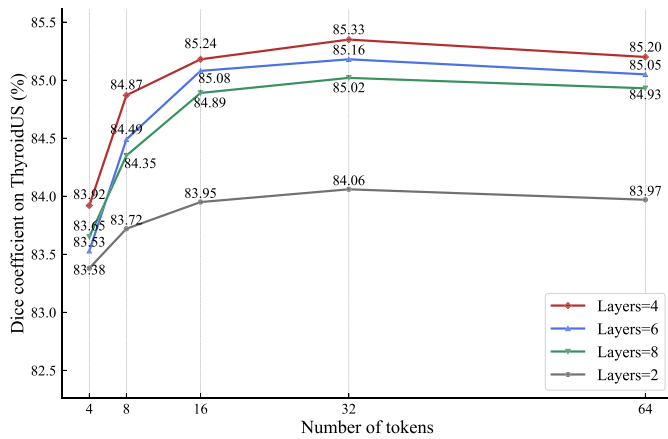


Fig. 9. Dice coefficient on ThyroidUS under different numbers of tokens and Transformer layers. When the number of tokens is 16, the accuracy does not improve significantly as the number of tokens increases. For the number of layers, the network performs best when is 4.

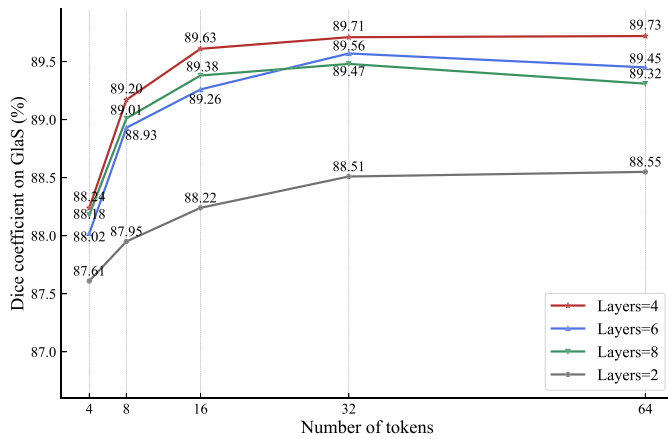


Fig. 10. Dice coefficient on GlaS under different numbers of tokens and Transformer layers. When the number of tokens is 16 and the number of layers is 4, the network gets better performance.

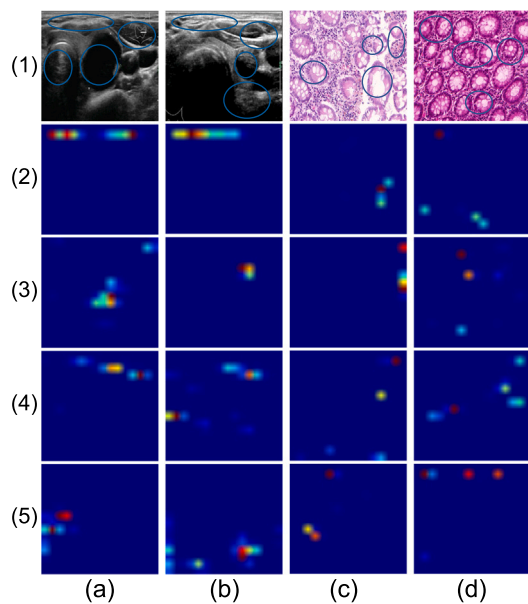


Fig. 11. Visualization of the area represented by tokens. ATEM separates tissues or structural features at different locations in the image and aggregates them as tokens to feed into Transformer.

Table 2

Comparison with average pooling on ThyroidUS and GlaS datasets in Dice coefficient.

	ThyroidUS	GlaS
Average pooling ($K = 16$)	84.51 _{$p=0.039$}	88.77 _{$p=0.015$}
Ours ($K = 16$)	85.24	89.63

the nodule can be accurately located. On the other hand, we also find that for the edge position of the nodule, the segmentation effect is not ideal, which needs further improvement. The reasons may be as follows. There are a large number of heterogeneous textures and folds in the edge region of thyroid nodules, and the semantics contained in these pixels are complex and fuzzy, which makes it difficult for the model to recognize these features, so the segmentation effect of nodule edges is not smooth enough.

Fig. 8 shows the segmentation effect on the GlaS and the MoNuSeg dataset. The first row shows the results on the GlaS dataset and the second row shows the results on the MoNuSeg dataset. According to the segmentation effect on GlaS dataset, ATTransUNet is more accurate in gland localization. This is consistent with the analysis in Fig. 7, where ATTransUNet better establishes the interrelation between the glands, and there are no mis-segmented regions. In terms of the segmentation effect on the MoNuSeg dataset, the proposed model has a good segmentation effect on cells in some areas, but does not have an obvious advantage in small and dense segmentation areas. In general, the proposed method is more accurate for the location of the segmented tissues or structure, and can better distinguish the features of different positions in the image. But the segmentation accuracy of the edge region needs to be further improved.

4.3. Ablation experiment

4.3.1. Ablation of the number of tokens and transformer layers

To find the appropriate model, we conducted an ablation experiment, which studied the number of tokens on the ATEM and the number of Transformer layers. The results on the ThyroidUS and GlaS datasets are presented in Figs. 9 and 10, respectively. To explore the impact of the number of tokens on the ATEM, we set the number of tokens as 4, 8, 16, 32, and 64, respectively. Similarly, To investigate the influence of the number of Transformer layers, we set the number of layers as 2, 4, 6, and 8, respectively. It is obvious that the highest Dice coefficient occurs when there are four Transformer layers in Figs. 9 and 10. It also can be seen from them that the worst performance happens when the number of Transformer layers is two. If we optimize the number of tokens, the results might change when the number of Transformer layers is four. As shown in Fig. 9, there is a trend of accuracy increasing for Dice when the number of tokens increases from 4 to 32. The performance decreases when the number of tokens is 64. Fig. 10 shows a similar trend. The ablation study on the number of tokens confirms the point of this paper: a few tokens can summarize the semantic information of the image, and there is no need to model all pixels in an image. To balance efficiency and accuracy, the number of tokens is set to 16 and the number of Transformer layers is set to 4.

Fig. 11 (1) represents the original image and Fig. 11 (2)–(5) visualize the area covered by the four different tokens that are generated by ATEM. The first two columns are visualizations on ThyroidUS dataset. From Fig. 11 (2)–(5), columns (a)–(b), it can be observed that the ATEM separates the features in the image into the skin, thyroid, skeletal muscle groups and trachea or artery. Further, ATEM will aggregate these features into tokens for self-attention modeling. The last two columns show the visualization of ATEM on GlaS dataset. In Fig. 11 (2)–(5), columns (c)–(d), it can be seen that ATEM separates the different glands. Fig. 11 illustrates from the perspective of visualization that ATEM can separate local tissues or structures in images, and aggregate them to establish long-range dependency.

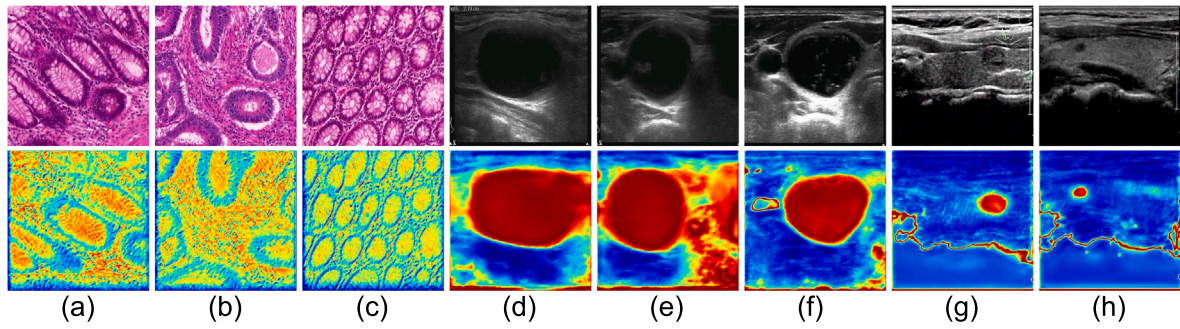


Fig. 12. Visualization of the weight map of the SFRM located in the last layer of the decoder. On histopathological images, SFRM enhanced areas such as glands and tissue fluid. In ultrasound images, SFRM mainly enhanced the thyroid and vascular regions.

Table 3
Ablation study of different feature reprojection methods on ThyroidUS and GlaS datasets in Dice coefficient.

	ThyroidUS	GlaS
Linear reprojection	84.71 _{p=0.057}	89.06 _{p=0.040}
Ours	85.24	89.63

Table 4
Ablation of the SFRM on ThyroidUS and GlaS datasets in Dice coefficient.

	ThyroidUS	GlaS
w/o SFRM	84.58 _{p=0.048}	88.79 _{p=0.014}
w/ Head	84.74 _{p=0.092}	89.08 _{p=0.043}
Spatial attention	85.06 _{p=0.248}	89.34 _{p=0.149}
Ours	85.24	89.63

Finally, we performed ablation experiments on the ThyroidUS and GlaS datasets by replacing ATEM with adaptive average pooling. Table 2 shows that, compared with adaptive pooling, ATEM with 16 tokens achieves higher segmentation accuracy and the Dice coefficient increases by 0.73% and 0.86%, respectively. It can also be inferred that, combined with the visual representation in Fig. 11, it can be inferred that the ATEM can better mine the irregular tissues or feature regions than the simple pooling method.

4.3.2. Ablation of the feature reprojection mechanism

Feature Reprojection Mechanism, as a supplement to the ATEM, aims to reproject the output of the tokens through Transformer to the spatial feature map for segmentation tasks. This approach can be naturally associated with the substitution of linear reprojection. Therefore, Table 3 shows the comparison between the linear reprojection and our method on Dice coefficient. Our method has achieved apparent improvement by 0.53% and 0.57%, respectively. It can be concluded that the Feature Reprojection Mechanism makes good use of the mutual query between tokens and pixels of the feature map, so that the recovered feature can better recognize the context.

4.3.3. Ablation of the selective feature reinforcement module

We introduce SFRM for the first and last convolution layer of the decoder. The SFRM is similar to the spatial attention [42] in nature. To better illustrate the advantages of this component, three ablation experiments on ThyroidUS and GlaS datasets are shown in Table 4: Compared with the results of not using SFRM, the introduction of this module significantly improves the accuracy, and the Dice coefficient on the two datasets is increased by 0.66% and 0.84%, respectively. The second row of the Table 4 shows that when SFRM is applied only to the head of the decoder, the Dice coefficient on both datasets is lower

Table 5
The check table for module ablations on ThyroidUS and GlaS datasets in Dice coefficient.

Tokens ($K = 16$) & Layers ($L = 4$)	Feature reprojection	SFRM	ThyroidUS	GlaS
			83.05 _{p=0.005}	87.26 _{p=0.006}
✓			84.18 _{p=0.016}	88.32 _{p=0.011}
✓	✓		84.58 _{p=0.048}	88.79 _{p=0.019}
✓	✓	✓	85.24	89.63

than the standard setting. We conclude that this is because the use of single-layer SFRM does not enable the network to enhance the features in the image from different levels. Finally, by replacing the SFRM with the spatial attention mechanism in the same location, our method has a small improvement compared. This phenomenon indicates that SFRM can be used as an alternative to the spatial attention mechanism, which can well locate and enhance the key tissue features.

In Fig. 12(a)–(h), the first row is the original images, and the second row visualizes the weight maps formed by the SFRM at the last layer of the decoder in the form of heat maps. Fig. 12(a)–(c) shows the heat maps of the SFRM on the GlaS dataset. According to Fig. 12(a)–(c), SFRM can better strengthen the area of glands and interstitium fluid, which can enable the network to better locate glands and complete the segmentation of glands. In Fig. 12(d)–(h), it was observed that SFRM enhanced the thyroid regions. In addition, in Fig. 12(e) we found that in addition to the enhancement of the nodule region to be segmented, SFRM also enhanced structures such as the right lobe or sternal muscle. We think that SFRM is essentially a method based on spatial attention. The structures outside these nodules belong to easily identifiable regions, and the impact of attention on such structures is weaker than the positive effect of network reinforcement on the regions to be segmented.

4.3.4. The check table for module ablations

Table 5 shows the variation of the Dice coefficient on the ThyroidUS and GlaS datasets when different modules are imposed on the network one by one. Compared to the benchmark without any modules, the Dice coefficient is significantly improved with the introduction of ATEM and Transformer (Based on the basic setup, the number of tokens is 16 and the number of Transformer layers is 4), so the accuracy is improved by 1.13% and 1.06% respectively. Then Feature Reprojection Mechanism and SFRM are applied to the network, and the segmentation performance steadily increases. Due to the introduction of the above modules proposed in this paper, the Dice coefficients on the two datasets are increased by 2.19% and 2.37%, respectively.

Through module ablation, We conclude that ATEM can mine and aggregate significant semantics in images and model long-range dependency of the semantics through Transformer. Furthermore, the

Feature Reprojection Mechanism restores the details of the feature map and integrates the context information of different regions. SFRM enhances the representation of key tissues and features in the image, thus resulting in higher segmentation accuracy.

5. Conclusion

Accurate and automatic segmentation of medical images is the key step in clinical diagnosis and analysis. In this paper, an enhanced Transformer hybrid segmentation network (ATTransUNet) is proposed to explore an efficient token mining method in medical images, and the self-attention mechanism is combined for medical image segmentation so as to achieve a balance between performance and efficiency. In addition, to further improve segmentation accuracy and integrate the respective advantages of CNN and Transformer, a Selective Feature Reinforcement Module (SFRM) is designed in this paper. The model proposed in this paper can accurately locate the structures to be segmented and has significantly improved the accuracy of medical image segmentation on multiple datasets. In terms of detection and classification of edge shapes, the model needs to be perfected. In the future, we will try to fuse features of different scales for more complex and diverse segmentation scenes to further improve the generalization and robustness of the model.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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